

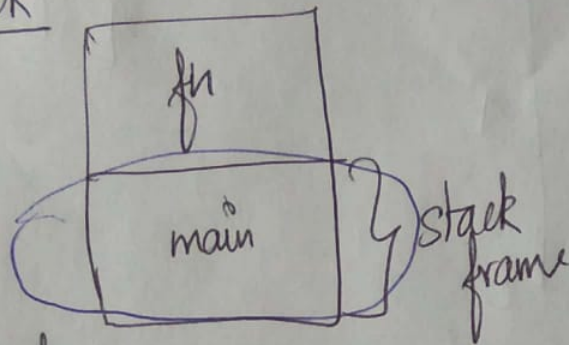
CS 354 - Lecture 4

Review

①

1. Stack vs Heap
2. Dynamic memory allocation $\leftarrow$ malloc()
free()
3. Linked Lists

Call Stack



main()

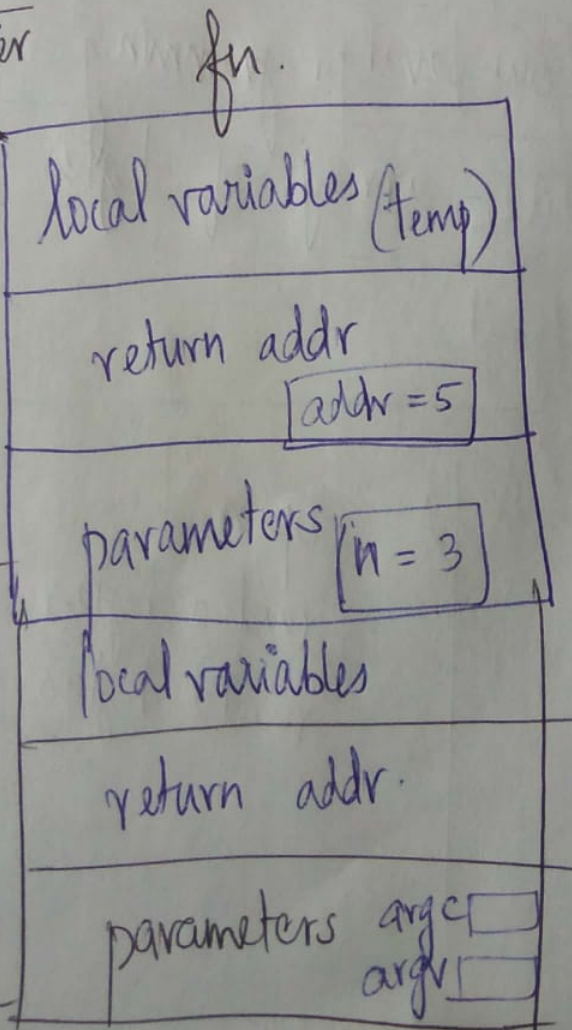
↓
fn(n)

int argc, char *argv

Top of the stack
stack pointer

stack
frame
for
fn.

main

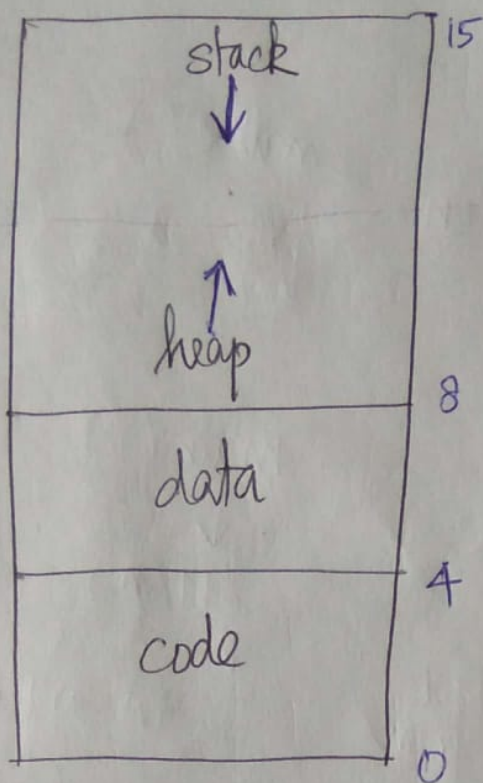


main() {
addr | 1 _____
| 2 _____
| 3 _____
| 4 fn(3);
| 5
| 6 _____
| }
}

fn(int n)
int temp;

}

(2)



Address Space of a program.

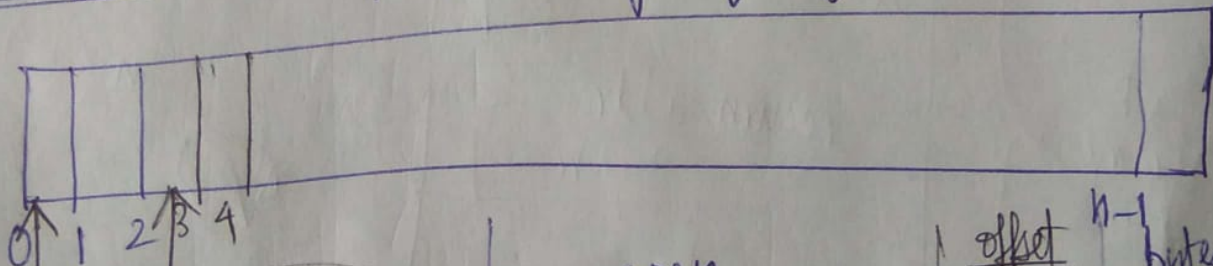
32-bits

$$\begin{aligned} \text{A.S.} &= 2^{32} \text{ bytes} \\ &= 2^2 \times 2^{30} \text{ bytes} \\ &= \underline{4\text{GB}} \end{aligned}$$

Low-Level programming

File I/O

File - array of bytes



open
read
write
close

	offset	bytes read
open	0	
read 2 bytes	2	0-1
read 2 bytes	4	2-3

Open?

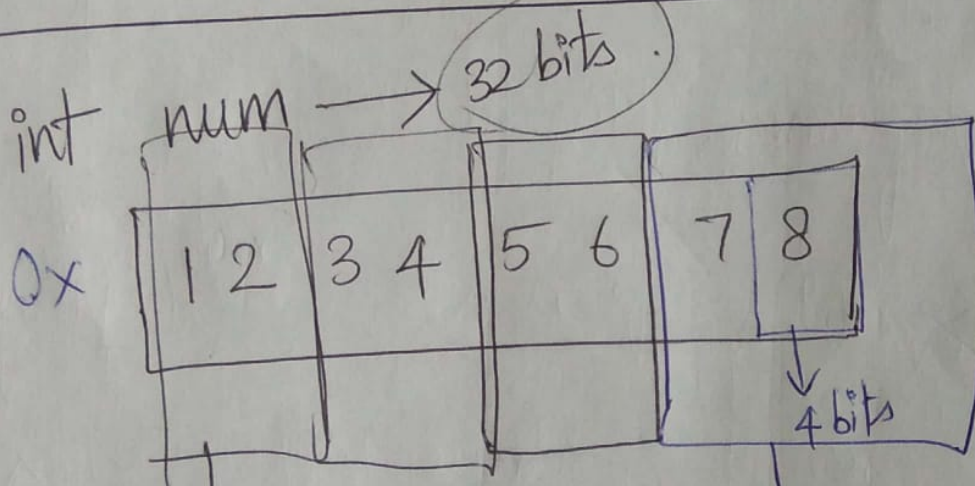
(3)

`fopen("test.txt", mode)`

text files
`fgets()`
`fputs()`

binary files
`fread()`
`fwrite()`

↓
r
w
a
"rb"



MSB
Big-endian

1 bytes
LSB
Little-endian

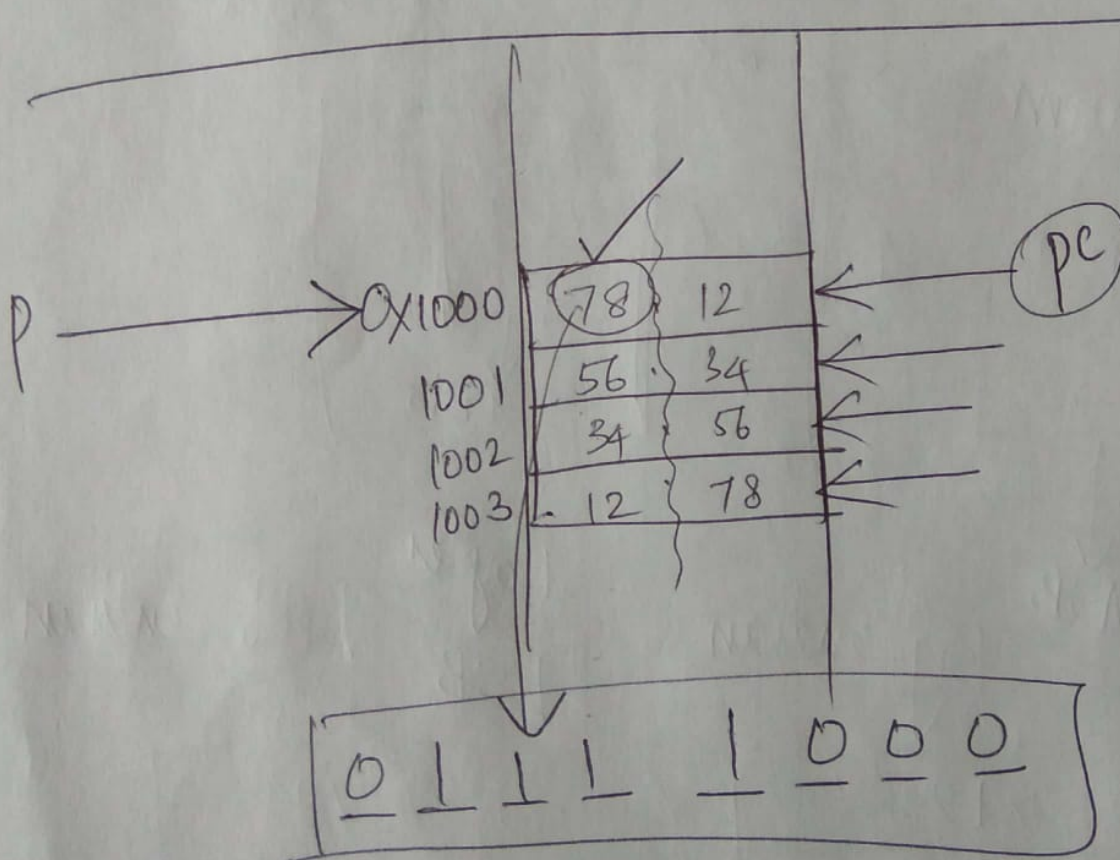
100

0	0x12
1	0x34
2	0x56
3	0x78

0	0x78
1	0x56
2	0x34
3	0x12



hton() → host to network
 ntohs() → network to host.



⑤
data types

char int float double

qualifiers: short long

(short int) x; // 2 bytes.

long int y;

(long long int) z; // 8 bytes

qualifier:

unsigned

0	0	0
0	1	1
1	0	2
1	1	3

signed

- - -
(-2 -1 0 +1)

3 bits - unsigned
- - -
0 to (7)^{2³-1}

unsigned int ui;

max → 2³²-1

signed

-4 to +3

int si;

-2³¹ to 2³¹-1

(32 bit)

$$\frac{1}{2^7} \quad \frac{1}{2^6} \quad \frac{1}{2^5} \quad \frac{1}{2^4} \quad \frac{1}{2^3} \quad \frac{1}{2^2} \quad \frac{1}{2^1} \quad \frac{1}{2^0}$$

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

Negative numbers

Signed
Magnitude
Form

0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

msb

lsb

+0

+1

+2

+3

-0

-1

-2

-3

msb

0 → +

1 → -

+2

-1

+1

010

101

111

-3

0	0	0	+0
0	0	1	+1
0	1	0	+2
0	1	1	+3
1	0	0	-3
1	0	1	-2
1	1	0	-1
1	1	1	-0

$$\sim 00_2 = (11)_2 = 3$$

$$\sim 01_2 = 10_2 = 2$$

ones' complement

2's complement

0	0	0	+0
0	0	1	+1
0	1	0	+2
0	1	1	+3
1	0	0	-4
1	0	1	-3
1	1	0	-2
1	1	1	-1

2's comp of 00_2

$$= \sim 00_2 + 1$$

$$= 11_2 + 1_2$$

$$= 100_2$$

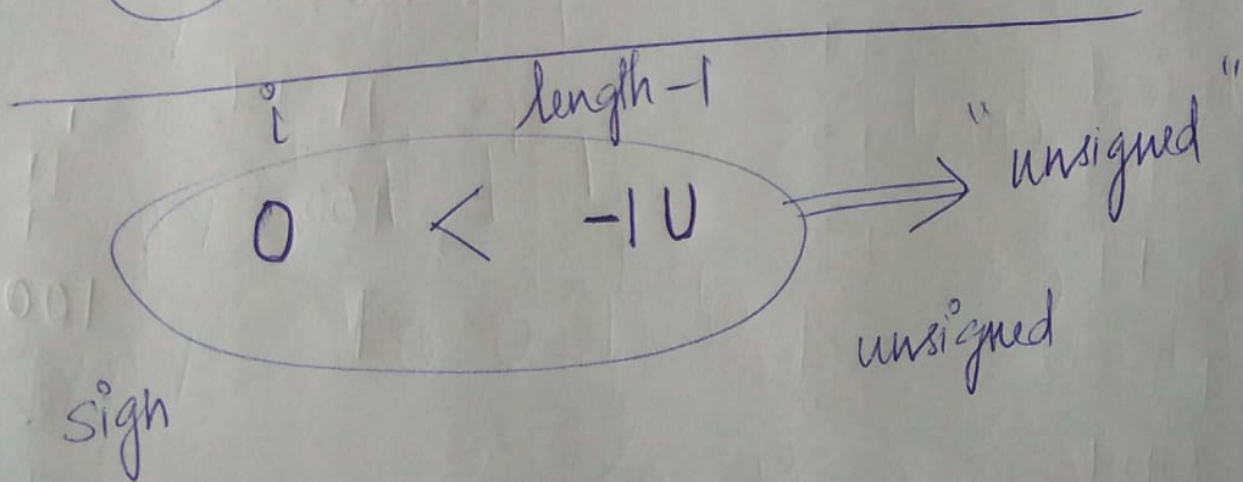
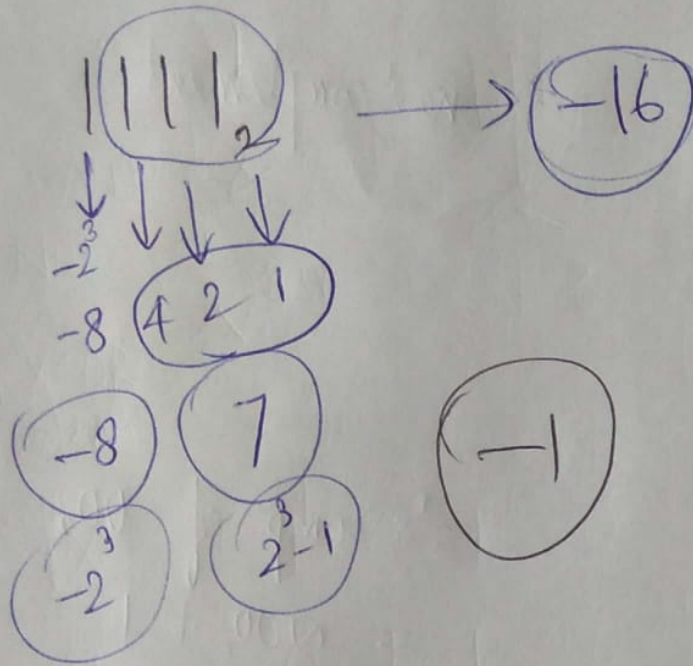
$$= 4$$

$$\begin{array}{r} 11 \\ \hline 100 \\ \hline \downarrow \\ 4 \end{array}$$

⑧

$$\begin{array}{c} 1 \ 0 \ 0 \ 1 \\ \downarrow \downarrow \downarrow \downarrow \\ 2^3 \ 2^2 \ 2^1 \ 2^0 \end{array} = -1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= -8 + 1 = \textcircled{-7}$$



0000

1111

⑨ Type conversions

short → int

char → short

short s;
int n = s;

0 1 1 0 → 0 0 0 0 0 1 1 0
↓ 6 ↓ 6

Sign extension

1 0 0 1 →

1 1 1 1 1 0 0 1

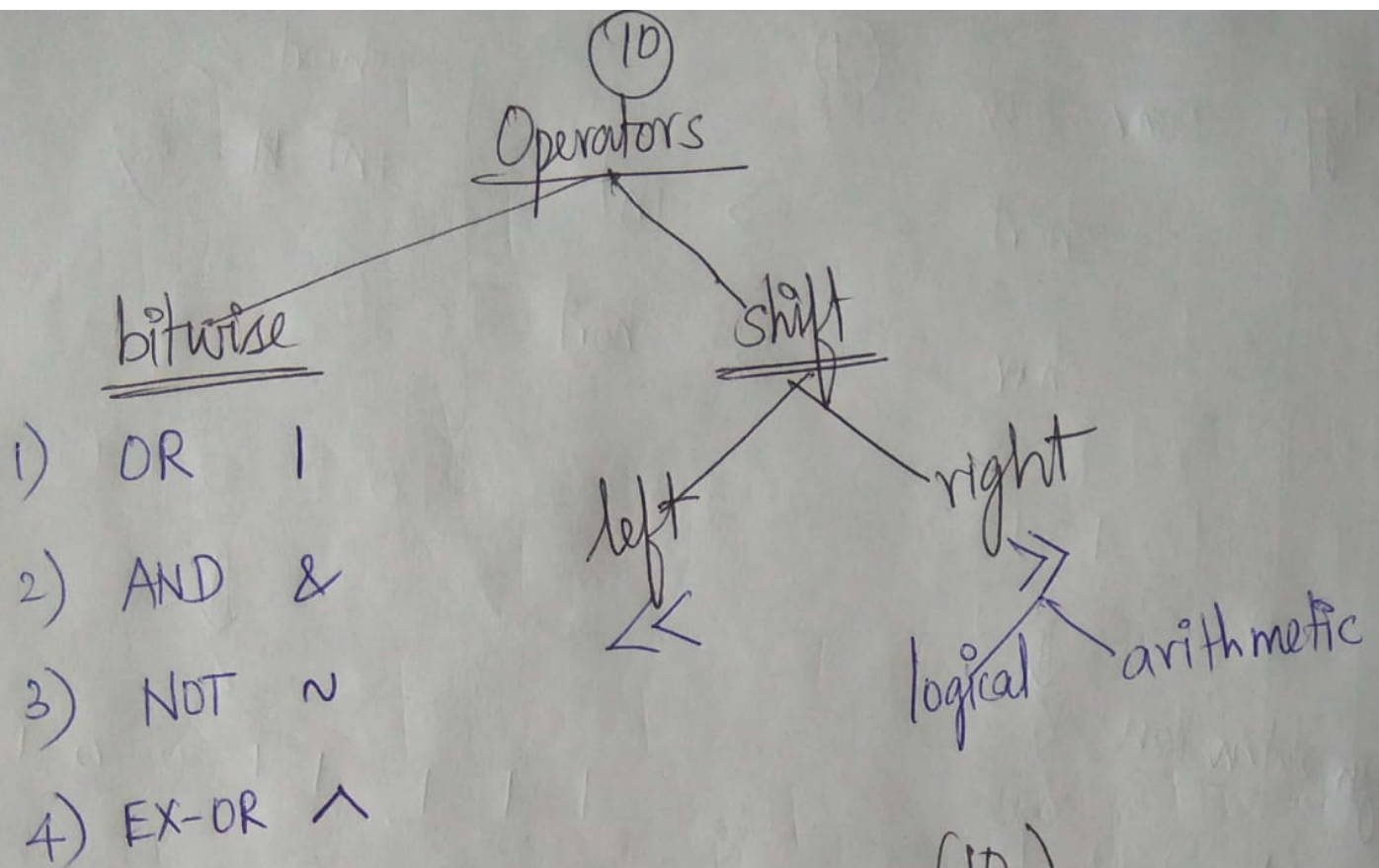
1 0 1 → 1 1 0 1

$$\begin{aligned} & -2^2 + 2^0 = -4 + 1 \\ & = -3 \end{aligned}$$

$$\begin{aligned} & -2^3 + 2^2 + 2^0 \\ & = -8 + 4 + 1 \\ & = -3 \end{aligned}$$

larger → smaller
1001 → 01

int n;
short s = n;
small ← big



$$X = 1010 \quad (10)$$

$$Y = 0110 \quad (6)$$

$$X | Y = 1110 \quad (14)$$

$$X \& Y = 0010 \quad (2)$$

$$\sim X = 0101 \quad (5)$$

$$X \wedge Y = 1100 \quad (12)$$

$$X = \boxed{1001} \textcircled{110010}$$

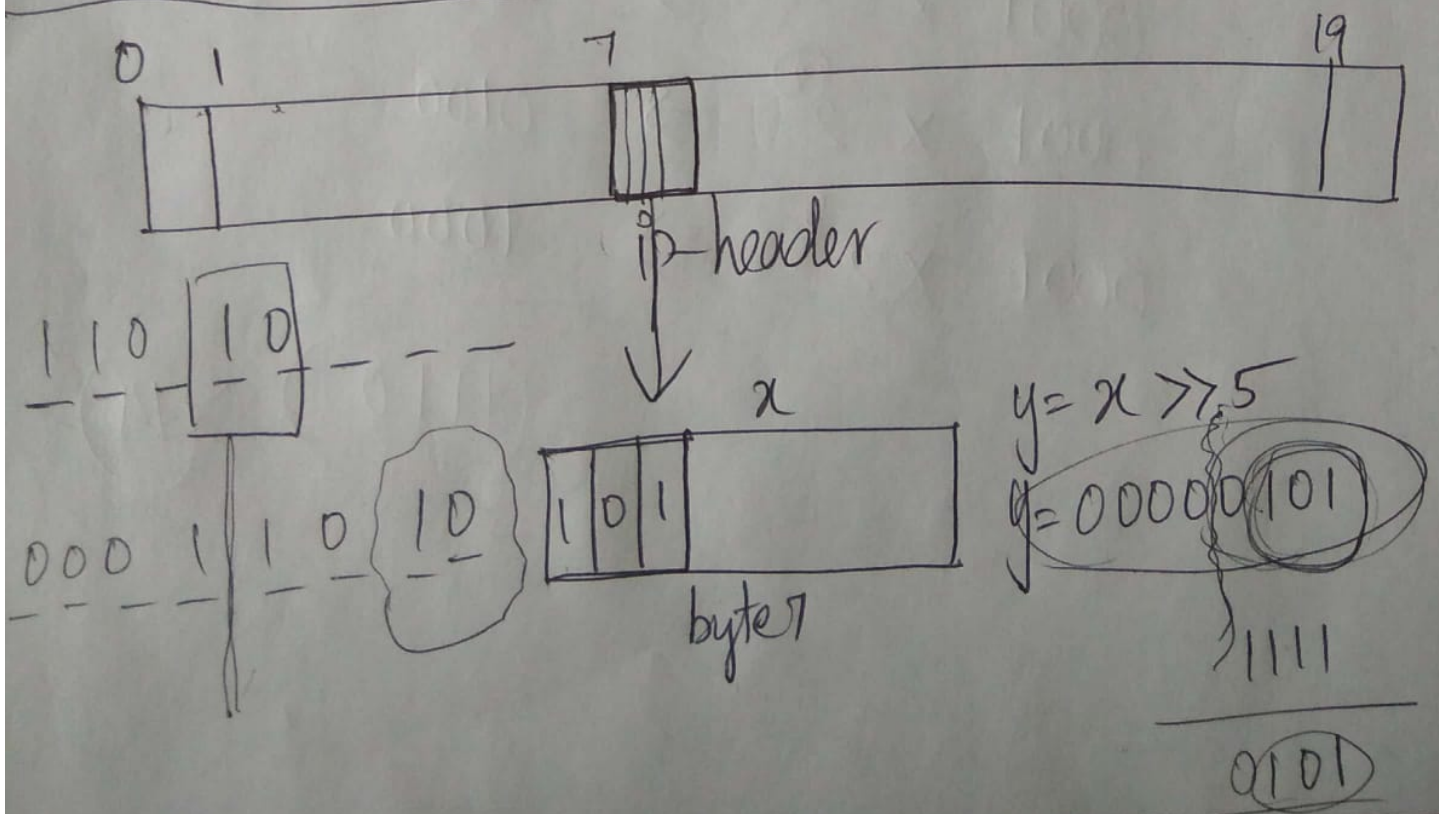
$$X \ll 4 = \underline{0} \underline{0} \underline{1} \underline{0} \underline{0} \underline{0} \underline{0} \underline{0}$$

$$X \gg 4 = \underline{0} \underline{0} \underline{0} \underline{0} \underline{1} \underline{0} \underline{0} \underline{1}$$

logical right shift
(unsigned numbers)

$$X \ggg 4 = \underline{1} \underline{1} \underline{1} \underline{1} \underline{1} \underline{0} \underline{0} \underline{1}$$

arithmetic right shift
(signed numbers)



$$\textcircled{14} \times 1$$

$$1110 \times 0001$$

(12)

$$\begin{array}{r} 1110 \\ 0001 \\ \hline 1110 \\ 0000 \end{array}$$

$$\begin{array}{c} 14 \\ \downarrow \\ 2 \end{array}$$

$$\left(2^3 + 2^2 + \textcircled{2}^1 \right) \times \textcircled{1}$$

0001

$$x * 2^K \equiv x \ll K$$

$$0001 \times 2^{\textcircled{1}} \Rightarrow 0010$$

$$0001 \times 2^{\textcircled{2}} \Rightarrow 0100$$

$$0001 \times 2^3 \Rightarrow 1000$$

$$\textcircled{1110} \Rightarrow \textcircled{14}$$